Mean Moderation Implementation Notes

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Purpose. This note records how the mean-moderation architecture (Architecture A) was implemented in `pmsimstats-ng`: three incorrect formulations, the calibrated formulation that was adopted, and a small pilot comparison of its power with the covariance-moderation architecture (Architecture B) under carryover. It is a quick-reference companion to the `.tex` version. *Review note (2026-10-07):* code claims were re-checked against the current `R/generateData.R`. The pilot power figures (50 replicates) have been superseded by paper 01 and by the closed-form results of `docs/37`. Dated notes mark these points below.

> **Note**: This is a quick-reference summary. For the full mathematical derivation, the positive-definiteness ceiling discussion, the `validateParameterGrid()` known limitation, and the empirical evaluation of the Hendrickson (2020) publication parameter grid under Architecture A, see the comprehensive document `19-mean-moderation-implementation-notes.tex` (rendered as `19-mean-moderation-implementation-notes.pdf`).

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Notation index and glossary

The notation follows `analysis/report/NOTATION.md`. *Canonical* symbols have the meaning given there. *Local* symbols are specific to this note. Code identifiers appear only in code spans. The package passes the moderation strength as `c.bm` under both architectures. In the mathematics it is β_{bm} under mean moderation and c_{bm} under covariance moderation. The pilot table reports both at the shared value $c_{bm} = \beta_{bm} = 0.45$ (NOTATION rule 2).

Table 1: Notation index: symbols used in this note

Symbol	Meaning	Status
BR	biologic (drug) response component	canonical
B	raw biomarker (code `bm`)	canonical
b	standardized biomarker $(B - \mu_{bm})/\sigma_{bm}$ (code `bm_z`)	canonical
μ_{bm}, σ_{bm}	biomarker mean and SD (124.33 and 15.36 in `extracted_bp`)	local (μ_{bm}); canonical (σ_{bm})
μ_{BR}, σ_{BR}	mean and SD of BR ($\sigma_{BR} = 8$ in `extracted_rp`)	local (μ_{BR}); canonical (σ_{BR})
β_{bm}	mean-moderation multiplier	canonical
c_{bm}	covariance-moderation correlation	canonical
$t_{1/2}$	carryover half-life, weeks	canonical
t_{sd}, t_{od}	time since discontinuation, time on drug (code `tsd`, `tod`)	canonical
N	total number of participants (convention of the pilot not re-verified)	canonical

Glossary. *Architecture A* is the former label for the mean-moderation architecture (`dgp_architecture = 'mean_moderation'`), and *Architecture B* the former label for the covariance-moderation architecture (`'mvn'`). *N-of-1* in the table is the Hendrickson Hybrid design. *Calibration* means choosing the mean-moderation shift so that a one-SD biomarker difference moves BR by the same amount as in the covariance-moderation conditional expectation.

Summary

This document records the implementation of Architecture A (direct mean moderation) for the biomarker-treatment interaction in `pmsimstats-ng`, including two failed approaches and the final correct formulation derived from the original 2025 codebase.

Background

The original R package code (`R/generateData.R`) uses Architecture B: the BM-BR interaction is encoded as a differential correlation in the covariance matrix of the joint MVN draw. Under carryover, this correlation erodes as the drug effect decays, producing the 40-60% power drops documented in `05-revision-summary.pdf`. (That file is no longer in `docs/`; flagged 2026-10-07. The 40-60% figure is a 50-replicate pilot result. See the note after the power table below.)

The 2025 tidyverse codebase (pre-alignment version in `~/prj/alz/01-pmsimstats/pmsimstats2025`, initial commit 957c8bd) used Architecture A: additive mean moderation applied post-MVN draw. That version showed full power preservation under carryover.

Implementation Attempts

Attempt 1: Multiplicative scaling (incorrect)

```
dat[, (br_col) := get(br_col) * (1 + beta_bm * bm_centered)]
```

Problem: When the base BR value is near zero (off-drug timepoints with minimal carryover), multiplying by `(1 + small)` produces negligible moderation signal. The analysis model sees strong biomarker signal on-drug but almost none off-drug, mimicking Architecture B's correlation erosion. Power dropped about 60% under carryover, no improvement over MVN.

Attempt 2: Additive with drug exposure decay (incorrect)

```
drug_exposure <- (1/2)^(tsd / t1half)  # on-drug=1, decays off-drug
dat[, (br_col) := get(br_col) + beta_bm * bm_centered *
  drug_exposure * br_sd]
```

Problem: Drug exposure decays rapidly during off-drug periods (e.g., 0.18 at first off-drug timepoint, 0.001 by the fourth, at 2.5-week spacing with $t_{1/2} = 1$; computed). The additive signal is proportional to this decay, so the off-drug moderation is still negligible relative to noise ($\sigma_{BR} = 8$; corrected 2026-10-07 from `br_sd = 5`, against `extracted_rp`). The `bm:Dbc` interaction term sees minimal contrast. Power remained low under carryover.

Attempt 3: Additive with time-on-drug (incorrect scaling)

```
dat[, (br_col) := get(br_col) + bm * tod * beta_bm]
```

Source: Original 2025 codebase, initial commit 957c8bd, `analysis/scripts/pm_functions.R` lines ~650-670:

```
treatment_status <- trial_design$tod[timepoint_idx]
participant_data[[br_col]] <- participant_data[[br_col]] +
  biomarker * treatment_status * interaction_strength
```

Problem: With uncentered biomarker (mean about 124) and t_{od} scaling with weeks (up to 10-20), the moderation shift at interaction strength 0.45 was about $124 \times 10 \times 0.45 = 558$ units at $t_{od} = 10$, and larger at longer exposure (the earlier "~650" corresponds to $t_{od} \approx 11.6$; made consistent with `docs/23` on 2026-10-07), against BR noise SD of 8. Power was 1.00 everywhere. This was an artifact of a grossly overpowered signal, not genuine resilience to carryover. The original 2025 codebase had this same scaling issue.

(Corrected 2026-10-08: the figure of about 558 units is the total shift at the mean biomarker, and almost all of it is biomarker-free drift, not moderation. Writing $B_i = \mu_{bm} + \sigma_{bm} b_i$, the shift is $\beta_{bm}\mu_{bm}t_{od} + \beta_{bm}\sigma_{bm}t_{od}\, b_i$. At $\beta_{bm} = 0.45$, $\mu_{bm} = 124.33$, $\sigma_{bm} = 15.36$ and $t_{od} = 10$ the drift is 559.5, which the drug and time terms absorb and which carries no interaction information; the moderation that the interaction term sees is 69.1 per SD of the biomarker, about 8.6 σ_{BR} (and 19.2 times the calibrated 3.6), growing with t_{od}. The comparison of 558 with the noise SD therefore overstates the signal-to-noise ratio by roughly a factor of 8; the conclusion, a grossly overpowered signal, stands. Verified by direct computation. See `analysis/report/compendium-mathematics.tex`, subsection "Historical mean-moderation formulations and the calibration check", Equation `eq:inva-attempt3`.)

Attempt 4: Centered, on-drug binary, calibrated (correct)

```
bm_z <- (dat$bm - bm_mean) / bm_sd
for (tp in 1:nP) {
  if (d[tp]$onDrug) {
    dat[, (br_col) := get(br_col) + c.bm * bm_z * br_sd]
  }
}
```

In mathematical form, the shift applied at every on-drug visit is $\beta_{bm}\, b\, \sigma_{BR}$, with β_{bm} passed in code as `c.bm`.

Why this works: At $\beta_{bm} = c_{bm}$ the formulation matches Architecture B's conditional expectation from the MVN model:

$$E[BR \mid B, \text{on drug}] = \mu_{BR} + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_{bm}} \cdot (B - \mu_{bm}) = \mu_{BR} + c_{bm}\, \sigma_{BR}\, b$$

The standardized biomarker b (code `bm_z`) centers the signal (mean moderation $= 0$ in the population). The σ_{BR} (code `br_sd`) scaling produces comparable effect sizes between architectures: a 1-SD biomarker shift produces $\beta_{bm}\sigma_{BR}$ units of BR shift, 3.6 at $\beta_{bm} = 0.45$ and $\sigma_{BR} = 8$. The binary on-drug indicator ensures moderation is present whenever drug is active and absent otherwise.

Calibration ($c_{bm} = \beta_{bm} = 0.45$, extracted parameters):

- Architecture A: 3.36 BR units per SD of biomarker
- Architecture B: 3.48 BR units per SD of biomarker
- Signal-to-noise ratio: about 0.65-0.9

Both empirical shifts are within sampling error of the theoretical 3.6 (the `.tex` version gives the 500-participant crossover draw behind them). The signal-to-noise definition behind "0.65-0.9" is not stated, and the range was not reproduced. The ratio of the per-SD shift to σ_{BR} is 0.45 (flagged 2026-10-07).

Power (N-of-1, $N = 70$, $c_{bm} = \beta_{bm} = 0.45$, 50 replicates):

Table 2: N-of-1 power by $t_{1/2}$ for Architecture A and Architecture B

$t_{1/2}$	Architecture A	Architecture B
0.0	0.74	0.82
0.5	0.68	0.64
1.0	0.72	0.50

Architecture A power is stable across carryover levels (variation is Monte Carlo noise at 50 replicates, MCSE about 0.065). Architecture B shows the characteristic 40% relative decline (0.82 to 0.50).

Note (2026-10-07). The 0.5 and 1.0 Architecture A cells disagree with `docs/21`, which reports 0.72 and 0.68 where this note (and its `.tex` version) report 0.68 and 0.72. The source run was not located, so the discrepancy is unresolved. Either ordering lies within Monte Carlo error of flat power. Whether $N = 70$ here is a total or a per-path count was not re-verified (NOTATION rule 4). Paper 01's later re-run at a matched total of $N = 70$ gives Hybrid carryover losses of 6.9 percent (mean moderation) and 15.4 percent (covariance moderation), as recorded in `analysis/report/NOTATION.md` (Part 2). `docs/37` derives the result that, under on-drug-only mean moderation, power does not depend on the half-life. It also shows that under graded covariance coupling power declines monotonically with the half-life.

Architectural Difference

The key distinction between Architectures A and B is where the interaction signal lives:

- **Architecture B (MVN):** Signal is in the second moment (covariance). The analysis model infers the interaction from conditional correlations between bm and BR. Carryover erodes the off-drug correlation, compressing the contrast.
- **Architecture A (mean moderation):** Signal is in the first moment (mean). The moderation is a fixed additive shift to BR during on-drug timepoints. Off-drug timepoints carry no moderation signal regardless of carryover state. The analysis model detects the interaction from the differential mean between high-biomarker and low-biomarker participants when on drug.

Implementation Details

The `dgp_architecture` parameter is threaded through three functions:

- `buildSigma()`: when `dgp_architecture = "mean_moderation"`, the BM-BR correlation block is omitted from the covariance matrix.
- `generateData()`: after the MVN draw, the additive moderation `c.bm * bm_z * br_sd` is applied to the BR column at every on-drug timepoint (`tod > 0`), where `bm_z` is the standardized biomarker and `br_sd` is the BR standard deviation from the response parameters.
- `generateSimulatedResults()`: passes the parameter to both the sigma cache and the per-paramset data generation.

Default is `"mvn"` to preserve backward compatibility with the Hendrickson et al. (2020) publication. All 67 existing tests passed under the default at the time of the merge.

Note (2026-10-07, inspected `R/generateData.R`). The package has since gained a third value, `dgp_architecture = "combined"` (paper 11). It applies the mean shift with weight `c.bm_a` and the covariance coupling with weight `c.bm_b`. A `moderation_scaling` argument was also added. Its default `"constant"` reproduces the shift above, and `"trajectory"` scales it by the on-drug BR mean curve, normalized to mean magnitude σ_{BR}. `validateParameterGrid()` is still not threaded. It calls `buildSigma()` with the default `"mvn"`, so under mean moderation it can report spurious PD failures (the known limitation described in the `.tex` version).

Files Modified

- `R/generateData.R`: `buildSigma()` and `generateData()`
- `R/generateSimulatedResults.R`: parameter threading

Status

Merged to `master` on 2026-04-08 in commit `5ed7b41` (the `feature/mean-moderation-dgp` branch). That commit is not present in the current repository's history, which begins on 2026-05-01, so this statement is unverified here. The same merge also introduced the `implementations/` directory with three parallel collections (`original`, `original-extended`, `tidyverse`); the active `R/` package directory mirrors `implementations/original-extended/`.